

Name:Yuvarani.M

Course:GenAI

Activity 1:10 AI Model Providers and Latest Models Released in the Last 12 Months

Model Provider	Model	Release Date	Knowledge Cutoff	Cost of Tokens (USD/1M input/output)	Context Window	Type	When to Use
OpenAI	GPT-5	Aug 2025	Sep2024	\$1.25/\$10	400K	Multimodal	Coding, reasoning, AI agents, test automation
Anthropic	Claude4 Opus	May 2025	Jan2025	\$15/\$75	200K	Multimodal	Long documents, code review
Google DeepMind	Gemini 2.5Pro	Mar 2025	Jan2025	\$1.25/\$10	1M	Multimodal	Research , long-context analysis
Meta	Llama4 Maverick	Apr 2025	Not published	Hosting dependent	Upto 1M	Multimodal	Self-hosted AI
xAI	Grok4	Jul 2025	Dec2025	Varies	256K	Multimodal	Real-time reasoning
Mistral AI	Mistral Medium 3	May 2025	Not published	Varies	128K	Text	Enterprise assistants
Cohere	CommandA	Mar 2025	Apr2025	Varies	256K	Text	RAG, Chatbots
Alibaba Cloud	Qwen3	Apr 2025	Not published	Open/API	256K	Multimodal	Coding, multilingual
DeepSeek	DeepSeek V3	Dec 2024	Not published	Low-cost	128K	Text	Programming, math
Amazon	Nova Pro	Dec 2024	Not published	Bedrock pricing	300K	Multimodal	Enterprise AI

Activity 2 : Understanding Transformer Models

A Professional Introduction to Transformer Architecture

Based on:

- IBM – Transformer Model Introduction
- An Introduction to Transformers
- Jay Alammar – The Illustrated Transformer

1. Introduction to Transformer Models

What is a Transformer Model?

A Transformer is a deep learning architecture designed to understand and generate human language. It was introduced in the research paper "**Attention Is All You Need**" in 2017.

Before Transformers, most language models depended on Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs). These models processed words sequentially, meaning they read one word at a time.

Transformers introduced a new approach called **Attention Mechanism**, which allows the model to look at all words in a sentence together and understand the relationship between them.

This innovation changed the field of Natural Language Processing (NLP) and became the foundation for modern Generative AI systems.

Examples of Transformer-based AI systems:

- ChatGPT
- BERT
- Google Gemini
- AI translation systems
- AI coding assistants

2. Evolution of Artificial Intelligence

The development of AI can be understood through the following progression:

Artificial Intelligence



Machine Learning



Deep Learning



Neural Networks



Transformer Models





Large Language Models (LLMs)

Artificial Intelligence (AI)

AI enables machines to perform tasks that normally require human intelligence.

Examples:

- Speech recognition
- Image recognition
- Decision making

Machine Learning (ML)

Machine Learning allows systems to learn patterns from data without being explicitly programmed.

Example: A system learns from thousands of emails to identify spam messages.

Deep Learning

Deep Learning uses neural networks inspired by the human brain.

It enables machines to understand:

- Text
- Images
- Speech
- Complex patterns

3. Why Were Transformers Needed?

Before Transformers, NLP models mainly used RNNs and LSTMs.

Limitations of Traditional Models

1. Sequential Processing

RNNs process information one word at a time.

Example:

Sentence:

"The customer service agent resolved the issue quickly."

Processing:

The → customer → service → agent → resolved → the → issue

The model cannot process all words simultaneously.

2. Difficulty Understanding Long Context

In long sentences, earlier information may be forgotten.

Example:

"The company launched a new product. After several months, the product received excellent customer feedback."

Traditional models may struggle to connect "product" with later information.

3. Slow Training

Because processing happens sequentially, training requires more time.

Transformer Solution

Transformers introduced:

- Parallel processing
- Attention mechanism
- Better context understanding
- Faster training
- Scalability for large AI models

4. Transformer Model Workflow

A Transformer converts human language into mathematical representations and produces meaningful output.

High-Level Flow

Input Sentence



Tokenization



Word Embeddings



Positional Encoding



Transformer Layers



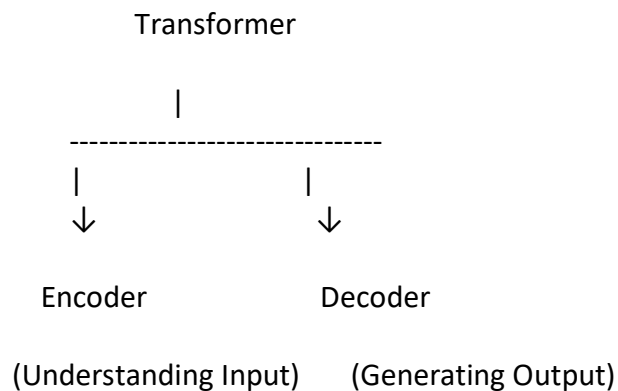
Prediction



Generated Output

5. Understanding Transformer Architecture

A Transformer mainly consists of two major components:



Encoder

The Encoder understands the input sentence.

It identifies:

- Meaning of words
- Relationship between words
- Context

The Encoder contains multiple layers.

Each layer consists of:

Encoder Layer



Multi-Head Attention





Add & Normalize



Feed Forward Network



Add & Normalize

Decoder

The Decoder generates the final output.

It uses:

- Previous output information
- Encoder information
- Attention mechanism

Decoder flow:

Previous Output



Masked Attention



Encoder Attention



Feed Forward Network



Final Prediction

6. Tokenization and Embeddings

Before a Transformer understands text, it converts words into numbers.

Tokenization

A sentence is divided into smaller units called tokens.

Example:

Sentence:

"AI improves customer experience"

Tokens:

AI | improves | customer | experience

Word Embeddings

Tokens are converted into numerical vectors.

Example:

Customer

↓

[0.25, 0.61, 0.87, 0.32]

These numbers represent the meaning of words.

7. Positional Encoding

Transformers process words together instead of one after another.

Because of this, the model needs information about word order.

Example:

Sentence 1:

"The dog chased the cat"

Sentence 2:

"The cat chased the dog"

Both contain the same words but have different meanings.

Positional Encoding helps the model understand the position of each word.

8. Attention Mechanism – The Heart of Transformers

The most important innovation in Transformers is the Attention Mechanism.

Attention allows the model to decide:

"Which words are important when understanding this word?"

Example:

Sentence:

"The bank approved my loan."

The word "bank" depends on surrounding words to understand whether it means:

- Financial institution
- River bank

Attention identifies the correct meaning using context.

Query, Key, and Value (QKV)

Attention works using three important concepts:

Query (Q)

Represents what information the model is looking for.

Example: "What meaning should I understand?"

Key (K)

Represents available information.

Example: "Which words are related?"

Value (V)

Contains the actual information passed forward.

Attention Flow

Input Words



Query
Key
Value



Similarity Calculation



Attention Score



Important Information Selected

9. Multi-Head Attention

A single attention mechanism may not capture all relationships.

Therefore, Transformers use multiple attention heads.

Example:

One head may focus on:

- Grammar relationship

Another head may focus on:

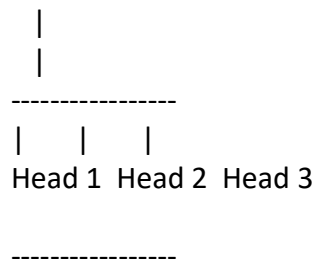
- Meaning relationship

Another head may focus on:

- Long-distance relationship

Flow:

Sentence



Combine Results



Final Representation

10. Feed Forward Network

After attention identifies important relationships, the Feed Forward Network processes the information further.

Its purpose:

- Improve understanding
- Learn complex patterns
- Generate better representations

11. Complete Transformer Architecture

INPUT TEXT



Token Embedding



Positional Encoding



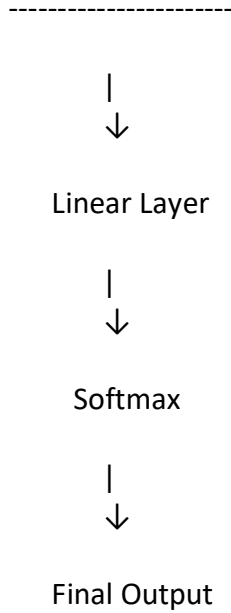
ENCODER STACK

Multi Head Attention
Feed Forward Network



DECODER STACK

Masked Attention
Encoder Attention
Feed Forward Network



12. Traditional Models vs Transformers

Feature	RNN/LSTM Models	Transformer Models
Processing	Sequential	Parallel
Training Speed	Slow	Faster
Context Understanding	Limited	Better
Long Sentences	Difficult	Effective
Scalability	Limited	Highly scalable
Used In	Earlier NLP systems	Modern AI systems

13. Real-World Applications of Transformers

1. Chatbots

Example:

- Customer service assistants
- Virtual agents

2. Translation

Transformers power:

- Language translation
- Multilingual AI systems

3. Content Generation

Applications:

- Email writing
- Article generation

- Summaries

4. Code Generation

AI assistants can generate:

- Programs
- Test scripts
- Documentation

5. Enterprise Applications

Examples:

- Customer support automation
- Knowledge search
- Document analysis
- AI-powered decision support

14. Key Takeaways

From IBM Transformer Model

- Transformers changed AI language processing.
- Attention is the core technology.
- They provide better scalability.
- They support modern Generative AI.

From Introduction to Transformers

- Text is converted into tokens.
- Tokens become embeddings.
- Positional encoding maintains word order.
- Encoder understands input.
- Decoder generates output.

From Illustrated Transformer

- Attention helps identify important relationships.
- Multi-head attention improves understanding.
- Encoder and Decoder work together.
- Transformer architecture supports modern AI systems.

Final Summary

Transformers represent one of the biggest breakthroughs in Artificial Intelligence. Unlike traditional models that process language sequentially, Transformers analyze relationships between words using attention mechanisms.

The ability to process information in parallel and understand context makes Transformers powerful and scalable. Today, Transformer architecture is the foundation behind Large Language Models (LLMs) such as GPT, BERT, and Gemini. Understanding Transformer architecture is essential for anyone working with Artificial Intelligence, Generative AI, automation, and modern software engineering.